

8 Proving Kepler's laws

In this lecture, we plan to derive Kepler's three laws.

Lecture 9: Kepler Orbits

Ph111 L9-1

- Derive Kepler's 3 laws
- M11 out today 4pm,
due Sunday 11:59pm

Important pitfall

Last time: $\mathcal{L} = \frac{1}{2}\mu(\dot{r}^2 + r^2\dot{\phi}^2)$
Don't take conserved quantities and plug them into Euler-Lagrange equations. That's because:
plugging in $\mathcal{L} = \mu r^2 \dot{\phi}$ enforces a required relationship between r and $\dot{\phi}$ so then you can't take partial derivatives anymore.

Pre-class video

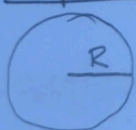
$r(t)$: not analytically solvable

$r(\phi)$: $\phi(r) - \phi(r_0) = \pm \frac{L}{\sqrt{2\mu}} \int_{r_0}^r \frac{dr'}{r'^2 \sqrt{H - U_{\text{eff}}(r')}}$
equation for shape

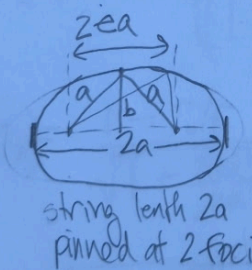
$\Rightarrow$ solvable, gives $\arcsin(\quad)$

Today: invert to get $r(\phi)$

Ellipses



circle

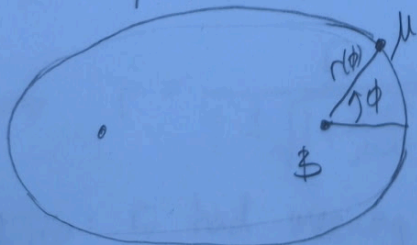


string length $2a$
pinned at 2 foci

$$b^2 + a^2 e^2 = a^2 \Rightarrow b = a\sqrt{1-e^2}$$

a = semimajor axis b = semiminor axis
 e = eccentricity, $e=0$ means circle
 $e=1$ means squished into line

In our equations:



$$r(\phi) = \frac{a(1-e^2)}{1+e\cos\phi}$$

$$r(\phi=0) = \frac{a(1+e)(1-e)}{1+e} = a(1-e) = r_{\text{min}}$$

"pericenter"

$$r(\phi=\pi) = \frac{a(1+e)(1-e)}{1-e} = a(1+e) = r_{\text{max}}$$

"apocenter"

We're going to run into something confusing later with that $\pm$:

$$r(\phi) = \frac{a(1-e^2)}{1 \pm e\cos\phi}$$

+ if $\phi=0$ at peri
- if $\phi=0$ at apo

since $\cos(\phi)$ is even $\rightarrow$

from end of pre-class video:

$$\phi(r) - \phi(r_0) = \pm \arcsin \left[\frac{\alpha r - \ell^2/\mu}{r\alpha(1 + 2H\ell^2/\mu\alpha^2)^{1/2}} \right]$$

$$= \pm \arcsin \left[\frac{1 - \frac{1}{r} \frac{\ell^2}{\mu\alpha}}{1 + \frac{2H\ell^2}{\mu\alpha^2}}^{1/2} \right]$$

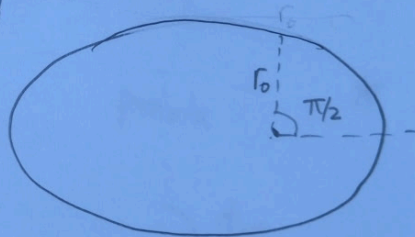
$$\mu = \frac{m_1 m_2}{m_1 + m_2}$$

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$$\alpha = G(m_1 + m_2)/r^2$$

Recall: $r_0 = \frac{\ell^2}{\mu\alpha}$

We need to decide where $\phi=0$. It would be nice to set $\phi(r_0)=0$. But there's a cosine in the ellipse equation and an arcsin here. So let's set $\phi(r_0) = \pi/2$ →



Then:

$$\phi(r) - \frac{\pi}{2} = \pm \arcsin \left(\frac{1 - r_0/r}{1 + \frac{2H\ell^2}{\mu\alpha^2}}^{1/2} \right)$$

$$\pm \sin(\phi(r) - \pi/2) = \frac{1 - r_0/r}{\sqrt{1 + \frac{2H\ell^2}{\mu\alpha^2}}}$$

$$\Rightarrow \pm \cos \phi(r) \sqrt{1 + \frac{2H\ell^2}{\mu\alpha^2}} = 1 - \frac{r_0}{r}$$

$$\Rightarrow r = \frac{r_0}{1 \pm \sqrt{1 + \frac{2H\ell^2}{\mu\alpha^2}} \cos \phi}$$

If $r_0 = a(1-e^2) = \frac{\ell^2}{\mu\alpha}$
and $e = \sqrt{1 + \frac{2H\ell^2}{\mu\alpha^2}}$
then it's the same

Compare:

$$r(\phi) = \frac{a(1-e^2)}{1 \pm e \cos \phi}$$

Elia: r_0 had meaning — r_0 that minimizes U_{eff}

Ryan: At r_0 , we'd be at circular orbit. Then $e=0$

Then $r=r_0$

This gives a mapping between conserved quantities and geometry

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$$r_0 = a(1-e^2) = \frac{l^2}{\mu \alpha}$$

$$e = \sqrt{1 + \frac{2Hl^2}{\mu \alpha^2}}$$

l, H set by initial conditions

"semi-latus rectum" "that won't be on the exam"

Present applet setting l and H first — more intuitive

To get $e=0$, need $H < 0$.

Can e be imaginary? $H = -\frac{GM\mu}{2a}$

IF $H > 0$ then $a < 0$
and $e > 1$

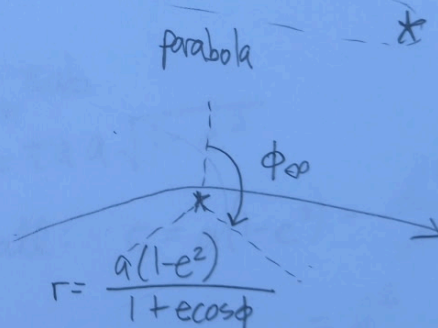
Long period comets have $H \approx 0$
so H goes way out and returns

3 types of solutions

$H > 0$ unbound — hyperbola
"way too much energy"

$H < 0$ bound — ellipse

$H = 0$ critically bound — parabola



start at $\phi = 0$ at r_{min} .
What needs to happen for $r \rightarrow \infty$?
Need $1 + e \cos \phi = 0$

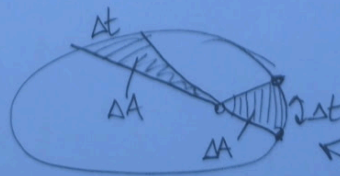
$$\Rightarrow -e \cos \phi = 1$$

$$\Rightarrow \cos \phi_0 = -\frac{1}{e} \quad \text{also:}$$

$$\Rightarrow e > 1 \quad a < 0$$

Kepler II

$$\frac{dA}{dt} = \text{const}$$



"sweeping out stumpy area over here"

Imagine a planet at distance r with tangential velocity $\vec{v}_t$
Wait dt



$$dA = \frac{1}{2} r^2 d\phi$$

$$r v_t \sin \theta = |\vec{r} \times \vec{v}_t|$$

so $dA = \frac{1}{2} r v \sin \theta dt = \frac{1}{2} |\vec{r} \times \vec{v}| dt$

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$$\Rightarrow \frac{dA}{dt} = \frac{1}{2} |\vec{r} \times \vec{v}| \frac{\mu}{\mu} = \frac{1}{2\mu} \mu |\vec{r} \times \vec{v}|$$

$$\frac{dA}{dt} = \frac{1}{2\mu} |\vec{r} \times \mu \vec{v}| = \frac{l}{2\mu} = \text{constant}$$

First found distance to moon using lunar eclipses
then found distance to sun ... Greeks?

Kepler III

$$P^2 \propto a^3$$

P is confusing — momentum **
P is the time to sweep out full area of ellipse.

$$P = \frac{A_{\text{ellipse}}}{dA/dt}$$

$$A_{\text{ellipse}} = \pi a b$$

$$= \pi a^2 \sqrt{1-e^2}$$

Recall: $r_0 = a(1-e^2) = \frac{l^2}{\mu \alpha}$

$$= \frac{l^2}{\mu G M \mu}$$

$$P = \frac{\pi a^2 \sqrt{1-e^2}}{l/2\mu}$$

$$P = \frac{2\pi a^2 \sqrt{1-e^2}}{(G M a (1-e^2))^{1/2}}$$

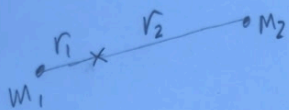
$$\Rightarrow l^2 = \mu^2 G M a (1-e^2)$$

$$\Rightarrow l = \mu \sqrt{G M a (1-e^2)}$$

$$P^2 = \frac{4\pi^2}{G M} a^3$$

$$\vec{r}_1, \vec{r}_2 \longleftrightarrow \vec{r}_{cm}, \vec{r}$$

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$$\frac{r_1}{r_2} = \frac{m_2}{m_1}$$

Talk about wobble of stars to find planets.

↳ Doppler shift $\frac{\Delta\lambda}{\lambda} = \frac{v_r}{c}$

Sun orbits at 10 cm/s

Need incredibly sensitive spectrometers to see $\frac{10 \text{ cm/s}}{3 \times 10^8 \text{ m/s}}$